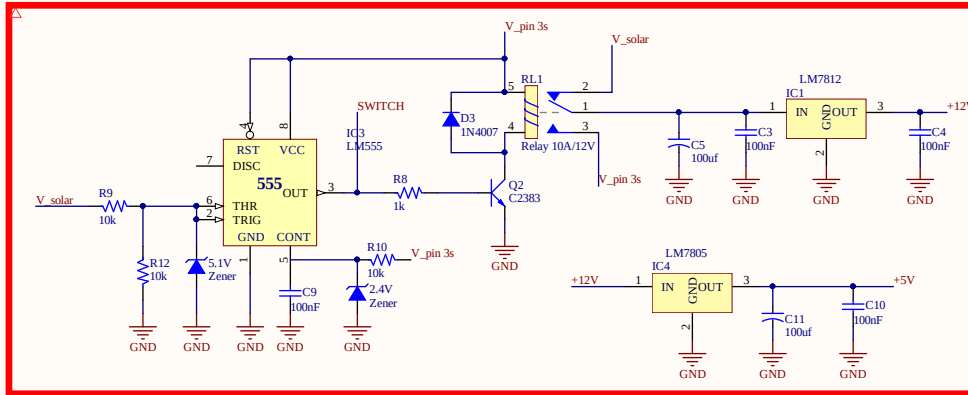
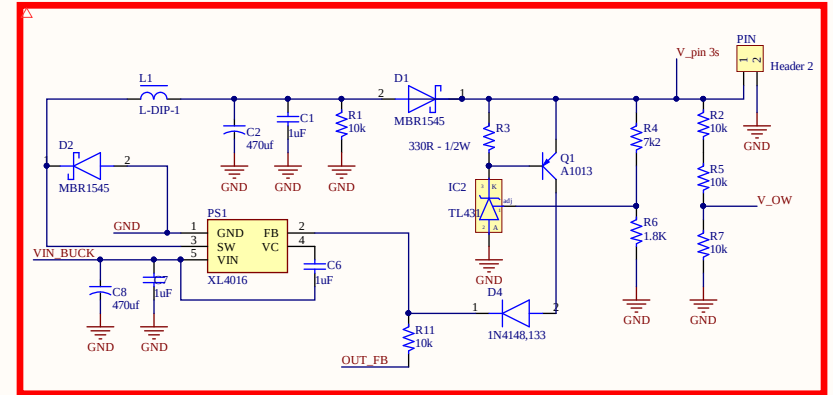


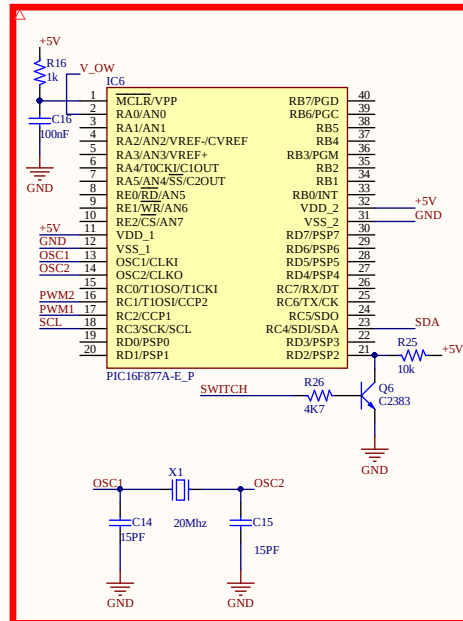
AUTO DAY/NIGHT RELAY SWITCH



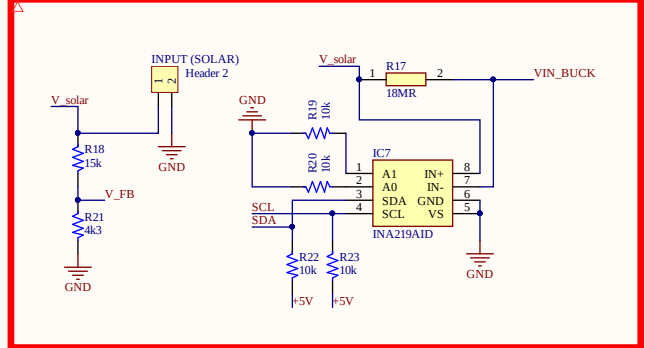
BUCK CONVERTER



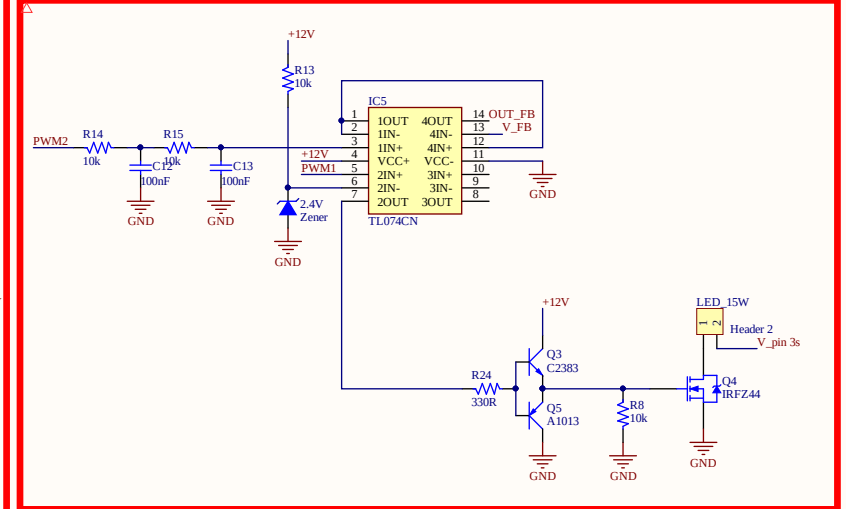
MAIN CONTROLLER



SOLAR INPUT & VOLTAGE / CURRENT SENSOR



GATE DRIVER & LOAD SWITCH



TITLE: SOLAR GARDEN LIGHT CONTROLLER

AUTHOR: VU VAN THI

REV: 1.0

DATE: 2026-09-21

Comment	Description	Designator	Footprint	LibF
Zener		2.4V, 5.1V	D1A	Zen
Cap Semi	Capacitor (Semicond	C1, C3, C4, C6, C7, C	C1206	Cap
470uf	Capacitor (Semicond	C2, C8	CA-ALU-6.3R	CN-
100uf	Capacitor (Semicond	C5, C11	CA-ALU-6.3R	CN-
MBR1545	Schottky Diode	D1, D2	TO508P498X1010X19	MBR
1N4007	Diode 1A	D3	D1A	D1A
1N4148,133	Diode	D4	DIOAD1165W56L425	1N4
LM7812	On ap +, from + 5 to	IC1	QTO-220	LM7
TL431	Programmable Shunt	IC2	TO92-123	TL4
LM555	Timer	IC3	DIP8/W0.3	LM5
LM7805	On ap +, from + 5 to	IC4	QTO-220	LM7
TL074CN	Integrated Circuit	IC5	DIP794W53P254L193	TL0
PIC16F877A-E_P	Integrated Circuit	IC6	DIP1550W56P254L52	PIC
INA219AID	Integrated Circuit	IC7	SOIC127P600X175-8	INA
Header 2	Header 2 (Green)	INPUT (SOLAR), LED	HED2	Hed
L-DIP-1	Inductor	L1	L-DIP-M-16.5x8.5	L-D
XL4016	Power Supply	PS1	XL4016	XL4
A1013	P 600mW, ECB	Q1, Q5	QTO-92	PNF
C2383	P 600mW, ECB	Q2, Q3, Q6	QTO-92	NPN
IRFZ44	HEXFET N-Channel P	Q4	QTO-220H3	IRF
10k	Resistor 1/4W	R1, R2, R5, R7, R8, R	SM-RES-1206	R
330R - 1/2W	Resistor 1/4W	R3	SM-RES-1206	R
7k2	Resistor 1/4W	R4	SM-RES-1206	R